For simplicity, we accept that the Actuator endpoints are available without authentication from within the cluster. Internal applications like Catalog Service are not affected, since their Actuator endpoints are not accessible through Spring Cloud Gateway. On the other hand, the Edge Service ones are currently accessible via the public internet.

That's not safe in a production environment. A simple way of fixing that is configuring the Ingress to block any request to the `/actuator/**` endpoints from outside the cluster. They will all still be available from within the cluster so that the health probes can work. We are using an NGINX-based Ingress Controller, so we can use its configuration language to express a *deny rule* for the Actuator endpoints.

In the source code repository accompanying the book, you can check the final results in the Chapter15/15-end folder (https://github.com/ThomasVitale/cloud-native-spring-in-action).

15.4 Continuous deployment with GitOps

Traditionally, continuous deployment is implemented by adding a further step to the production stage of the deployment pipeline. This additional step would authenticate with the target platform (such as a virtual machine or a Kubernetes cluster) and deploy the new version of the application. In recent years, a different approach has become more and more popular: GitOps. The term was coined by Alexis Richardson, CEO and founder of Weaveworks (www.weave.works).

GitOps is a set of practices for operating and managing software systems, enabling continuous delivery and deployment while ensuring agility and reliability. Compared to the traditional approach, GitOps favors decoupling between delivery and deployment. Instead of having the pipeline *pushing* deployments to the platform, it's the platform itself *pulling* the desired state from a source repository and performing deployments. In the first case, the deployment step is implemented within the production stage workflow. In the second case, which will be our focus, the deployment is still theoretically considered part of the production stage, but the implementation differs.